# A1 Q3b

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b) No, if $X_t$ is weakly stationary, $X_t^2$ is not necessarily weakly stationary.

As a counterexample, consider the time series from Q2b) once more, along with the definitions of $Z_0 \sim \mathcal{N}(0,1)$ and $Z_i \in \{-1, 1\}$ w/ prob. 0.5 each from Q3a).

For even $t$, when $X_t = Z_0 s_t$, note that $Var(X_t^2) = 2$ b/c $Var(Z_0^2) = 2\sigma^4$ (according to Math Stack Exchange Question 620045)

$= 2(1)$

$= 2$

For odd $t$, when $X_t = Z_i s_{t-1}$, $Var(X_t^2) = 0$ b/c the probability of this simple discrete distribution doesn't change.

Given that these variances differ based on when $t$ is even or odd, we have shown that the Covariance of $X_t^2$ relies on $t$. This violates the definition of weak stationarity.

Therefore we have shown a counterexample showing that $X_t$ may be weakly stationary, but $X_t^2$ is not weakly stationary.